

Naming Conventions

Consistent naming is required throughout the project.

3.1 Variables

Use `snake_case`.

```
file_path  
process_name  
command_result
```

Avoid:

```
filePath  
ProcessName  
f
```

unless a very short local variable is appropriate.

3.2 Functions

Use `snake_case` and start with a descriptive verb where appropriate.

```
create_folder()  
open_application()  
get_system_info()  
execute_command()
```

3.3 Classes

Use `PascalCase`.

```
FileManager  
CommandParser  
ExecutionEngine  
PluginManager  
SystemMonitor
```

3.4 Constants

Use uppercase `SNAKE_CASE`.

```
MAX_RETRIES  
DEFAULT_TIMEOUT  
PLUGIN_DIRECTORY
```

3.5 Modules and Files

Use lowercase `snake_case`.

```
command_parser.py  
execution_engine.py  
file_manager.py  
system_monitor.py
```

3.6 Boolean Variables

Boolean names should clearly indicate a true/false condition.

Preferred:

```
is_running  
is_enabled  
has_permission  
file_exists
```

Avoid ambiguous names such as:

```
status  
flag  
value
```

when they represent a boolean.

3.7 Plugin Names

Plugin names should clearly identify the functionality they provide.

Examples:

```
file_manager  
system_utils  
process_manager  
browser_plugin
```

Avoid generic names such as:

```
plugin1  
test_plugin  
manager  
helper
```

unless they are genuinely appropriate.